The Supply Side of HANK

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Motivation

- Does HA matter for the supply side of HANK models?
- A common assumption is that aggregate labor supply depends only on the marginal utility of average consumption. → As in RANK
- Not empirically supported:
  - Golosov et al. (2023)
  - Cantore et al. (2025) and Graves, Huckfeldt, and Swanson (2023)
- We look at the implications for labor supply, marginal costs, inflation dynamics, and monetary policy tradeoffs.

In this paper

- Study the distortions introduced by HA on the supply side of a general class of HANK models.
- Distortions captured by the sufficient-statistic cross-sectional wedge $\mathcal{Q}_t$: function of heterogeneity in consumption and productivity.
- Characterize an analytical decomposition (T-HANK) of $\mathcal{Q}_t$ and identify the two key relevant margins: intensive and extensive margins of heterogeneity.
- Unfavourable idiosyncratic risk shocks have independent recessionary demand-side, deflationary supply-side, and worsening distributional effects.

  becoming & remaining HtM

- Policy implications: no divine coincidence and strict inflation targeting is sub-optimal.

literature

HANK Model

McKay, Nakamura, and Steinsson (2016) / one asset version of Kaplan, Moll, and Violante (2018)

$$V_t(b_{h,t}, z_{h,t}) = \max_{c_{h,t},\, b_{h,t+1},\, n_{h,t}} \left\{ \frac{c_{h,t}^{1-\sigma}}{1-\sigma} - \frac{n_{h,t}^{1+\varphi}}{1+\varphi} + \beta \sum_{z_{h,t+1}} \Pr(z_{h,t+1} \mid z_{h,t})\, V_{t+1}(b_{h,t+1}, z_{h,t+1}) \right\}$$

subject to

$$c_{h,t} + b_{h,t+1} = (1+r_t) b_{h,t-1} + W_t z_{h,t}\, n_{h,t} - \tau_t \bar{\tau}(z_{h,t}) + d_t \bar{d}(z_{h,t})$$

$$b_{h,t+1} \geq \underline{b}.$$

- idiosyncratic labour productivity $z_{h,t}$ follows a Markov chain with transition matrix $\Pr(z_{h,t+1} | z_{h,t})$,
- stationary distribution $\mathcal{F}^z(z)$, with $E_{h,t} \equiv \int z_{h,t}\, d\mathcal{F}^z(z) = 1$,
- taxes and dividends proportional to $z_{h,t}$.

Individual Labor Supply

- Individual labor supply:

$$n_{h,t} = \left(W_t z_{h,t}\right)^{1/\varphi} c_{h,t}^{-\sigma/\varphi} \tag{1}$$

- Aggregate effective labor supply:

$$N_t = \int z_{h,t} n_{h,t} d\mathcal{F}_t(b,z) \tag{2}$$

- The cross-sectional distribution of productivity $\mathcal{F}^z(z)$ is constant, while the joint distribution $\mathcal{F}_t(b,z)$ is time-varying.
- Substituting (1) into (2) yields:

$$N_t = W_t^{1/\varphi} \underbrace{\int z_{h,t}^{(1+\varphi)/\varphi} c_{h,t}^{-\sigma/\varphi} d\mathcal{F}_t(b,z)}_{\mathcal{K}_t\text{: cross-sectional moment}=C_t^{-\sigma/\varphi}\text{ in RANK}} . \tag{3}$$

The $\mathcal{Q}$ Wedge

$$N_t = W_t^{1/\varphi} C_t^{-\sigma/\varphi} \int z_{h,t}^{(1+\varphi)/\varphi} \left(\frac{c_{h,t}}{C_t}\right)^{-\sigma/\varphi} d\mathcal{F}_t(b,z) = N_t^R \mathcal{Q}_t^{-1/\varphi}, \tag{4}$$

$$N_t^R = W_t^{1/\varphi} C_t^{-\sigma/\varphi} \tag{5}$$

$$\mathcal{Q}_t = \left[\int z_{h,t}^{(1+\varphi)/\varphi} \left(\frac{c_{h,t}}{C_t}\right)^{-\sigma/\varphi} d\mathcal{F}_t(b,z)\right]^{-\varphi} = \mathcal{K}_t^{-\varphi} C_t^{-\sigma}. \tag{6}$$

- $\mathcal{Q}_t$ is the ratio between the marginal utility of aggregate consumption and the average, productivity–adjusted marginal utility of individual consumption.
- $\mathcal{Q}_t = 1$ in RANK and "Canonical" HANK (Auclert, Rognlie, and Straub, 2025).

$\mathcal{Q}_t$ Decomposition

two sources of heterogeneity

- The composite wedge can be split into a productivity term, a consumption-inequality term, and a covariance term:

$$\mathcal{Q}_t^{-1/\varphi} = \underbrace{\mathcal{F}^z}_{E_h[z_{h,t}^{(1+\varphi)/\varphi}]} \cdot \underbrace{Q_{c,t}}_{E_{h,t}\left[\left(c_{h,t}/C_t\right)^{-\sigma/\varphi}\right]} + \underbrace{\mathcal{C}_t}_{\mathrm{Cov}_{h,t}\left(z_{h,t}^{(1+\varphi)/\varphi}, \left(c_{h,t}/C_t\right)^{-\sigma/\varphi}\right)} \tag{7}$$

- Productivity component: $\mathcal{F}^z \geq 1$
- Consumption-inequality component: $Q_{c,t} \geq 1$
- Covariance component: $\mathcal{C}_t < 0$
  - More productive households tend to consume more, so their relative marginal utility of consumption is lower.
- Both $\mathcal{F}^z$ and $Q_{c,t}$ increase with cross-sectional dispersion in productivity and consumption, respectively.

$\mathcal{Q}_t \leq 1$

- $\mathcal{C}_t < 0$ offsets the positive product $\mathcal{F}^z Q_{c,t}$, making the net wedge smaller.
- A necessary & sufficient condition for $\mathcal{Q}_t \leq 1$ ($\mathcal{Q}_t^{-1/\varphi} \geq 1$) is that the covariance is not too large in absolute value:

$$-\mathcal{C}_t \leq \mathcal{F}^z Q_{c,t} - 1.$$

- Under the assumption that $z_{h,t}$ and $c_{h,t}$ are jointly log-normally distributed in the cross-section, we can characterize $\mathcal{Q}_t$ as:

$$\mathcal{Q}_t = \exp\left(-\frac{1+\varphi}{2\varphi}\sigma_z^2 - \frac{\sigma(\sigma+\varphi)}{2\varphi}\sigma_{c,t}^2 + \frac{(1+\varphi)\sigma}{\varphi}\rho_{zc,t}\sigma_z\sigma_{c,t}\right), \tag{8}$$

- For empirically relevant calibrations (Heathcote, Storesletten, and Violante (2014)), and always when $\sigma = \varphi = 1$, we get $\mathcal{Q}_t \leq 1$.

Real Marginal costs and the NKPC

- Standard production sector using a linear technology and subject to Rotemberg pricing.
- Real marginal costs:

$$MC_t = exp(-a_t)W_t = \frac{Y_t^{\varphi+\sigma}}{\exp((1+\varphi)a_t)}\mathcal{Q}_t = MC_t^R\mathcal{Q}_t \tag{9}$$

- If $\mathcal{Q}_t \leq 1$, the real marginal cost schedule in HANK lies everywhere below the RANK schedule:
  - The HANK economy produces a given unit of output at lower marginal cost than the corresponding RANK economy.
  - The aggregate supply curve is therefore shifted outward by inequality, with the degree of the outward shift governed by $\mathcal{Q}_t$.
- More on HANK model

Real Marginal costs and the NKPC

- Log-linearise (9) around the zero-inflation, zero-output-gap steady state:

$$\widehat{mc}_t = (\varphi + \sigma)x_t + \hat{q}_t, \tag{10}$$

- The NKPC:

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{\kappa}{\sigma + \varphi}\widehat{mc}_t \tag{11}$$

- The Phillips curve in HANK acquires an additional endogenous term $\kappa(\sigma + \varphi)^{-1}\hat{q}_t$.
- A rise in inequality (lower $\hat{q}_t$) suppresses inflationary pressure at a given output gap: the outward shift of the AS curve reduces inflationary pressure.

A Tractable HANK

Bilbiie (2025)

- $\mathcal{Q}_t^{-1/\varphi} = \mathcal{F}^z Q_{c,t} + \mathcal{C}_t$
  - $\frac{c_h}{C}$: per capita variation in consumption inequality (intensive margin)
  - $\mathcal{F}_t(b, z)$: cross sectional distribution of agents (extensive margin)

- We build a T-HANK that captures these two relevant margins in an analytically tractable way.
  - Two household types (islands): savers (S) and hand-to-mouth households (H).
  - Homogeneous productivities.
  - Zero liquidity.
  - Limited risk sharing: within but not across types.
  - Utility function $U(C_j, N_j) \equiv exp(\xi_t)\frac{C_j^{1-\sigma}}{1-\sigma} - \frac{N_j^{1+\varphi}}{1+\varphi}$ for each agent type $j = S, H$.
  - Calvo pricing and flexible wages.

Details

Idiosyncratic state & endogenous HtM share

- 2 states Markov chain with stochastic and cyclical transition probabilities.

$$
\begin{aligned}
p_{S,t} &= p_S \exp(\eta_S \hat{y}_t + e_{S,t}), \\
p_{H,t} &= p_H \exp(-\eta_H \hat{y}_t + e_{H,t}), \\
\lambda_t &= p_{H,t}\lambda_{t-1} + (1 - p_{S,t})(1 - \lambda_{t-1}).
\end{aligned}
$$

- This relaxes the fixed-share restriction used in Bilbiie (2025) such that $\lambda_t = \lambda = \frac{1-p_S}{2-p_S-p_H}$ for each t.
- empirical evidence

2 Agents $\mathcal{Q}_t$ Wedge

1 No idiosyncratic productivity heterogeneity $\rightarrow \mathcal{F}^z = 1$ and $\mathcal{C}_t = 0$.

* Set $z_{h,t} = 1$ for all h and t. Then

$$\mathcal{Q}_t^{-1/\varphi} = \int \left(\frac{c_{h,t}}{C_t}\right)^{-\sigma/\varphi} d\mathcal{F}_t(b) \equiv Q_{c,t}, \tag{12}$$

2 Zero liquidity / Two types with limited risk sharing

* Under a *linear aggregator of hours in production*, we get

$$Q_{\text{lin},t}^{-1/\varphi} = (1-\lambda_t)\left(\frac{C_{S,t}}{C_t}\right)^{-\sigma/\varphi} + \lambda_t\left(\frac{C_{H,t}}{C_t}\right)^{-\sigma/\varphi} \tag{13}$$

3 Cobb-Douglas labor aggregator $N_t = N_{S,t}^{1-\lambda_t} N_{H,t}^{\lambda_t} \rightarrow W_{S,t}N_{S,t} = W_{H,t}N_{H,t} = W_t N_t$:

*

$$Q_t^{-1/\varphi} = \left[\left(\frac{C_{S,t}}{C_t}\right)^{-\sigma/\varphi}\right]^{1-\lambda_t}\left[\left(\frac{C_{H,t}}{C_t}\right)^{-\sigma/\varphi}\right]^{\lambda_t} \tag{14}$$

Aggregate Labor Supply Wedge

- Q_t is the ratio between the *marginal utility of the average consumption* and the *average of the marginal utilities of consumption*.

$$Q_t \equiv \left[\left(\frac{C_{S,t}}{C_t} \right)^{1-\lambda_t} \left(\frac{C_{H,t}}{C_t} \right)^{\lambda_t} \right]^{\sigma} = \left(\frac{\Gamma_t^{\,1-\lambda_t}}{\lambda_t + (1-\lambda_t)\Gamma_t} \right)^{\sigma} \tag{15}$$

- Intensive margin: changes in inequality $\Gamma_t \equiv \frac{C_{S,t}}{C_{H,t}}$
- Extensive margin: changes in agent shares λ_t
- Q_t measures the deviation from the frictionless marginal rate of substitution.

Aggregate Labor Supply Wedge

$$
\begin{aligned}
(1-\tau^w)W_t &= \overbrace{\exp(-\xi_t)\left(N_{S,t}^{\varphi}C_{S,t}^{\sigma}\right)^{1-\lambda_t}\left(N_{H,t}^{\varphi}C_{H,t}^{\sigma}\right)^{\lambda_t}}^{\text{Average } MRS_t} && (16)\\
&= \underbrace{\exp(-\xi_t)N_t^{\varphi}C_t^{\sigma}}_{\text{Frictionless } MRS_t^*} \cdot \underbrace{\left[\left(\frac{C_{S,t}}{C_t}\right)^{1-\lambda_t}\left(\frac{C_{H,t}}{C_t}\right)^{\lambda_t}\right]^{\sigma}}_{Q_t} && (17)
\end{aligned}
$$

First-Order Approximation (FOA): Supply Side

Steady state | Full model

no profits redistribution $\tau^D = 0$

$$\hat{c}_{S,t} = E_t\hat{c}_{S,t+1} - (1-p)E_t\hat{\gamma}_{t+1} - \frac{1}{\sigma}\left(\hat{\imath}_t - E_t\hat{\pi}_{t+1} + E_t\Delta\hat{\xi}_{t+1}\right) + p\frac{\Gamma^\sigma - 1}{\sigma}E_t\hat{p}_{S,t+1} \tag{18}$$

$$\hat{c}_{H,t} = \hat{w}_t + \hat{n}_t \tag{19}$$

$$\hat{q}_t = -\sigma(\lambda - \psi)\hat{\gamma}_t - \sigma(\lambda\gamma - \psi g)\hat{\lambda}_t \tag{20}$$

$$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} - \eta\hat{y}_t + p_H e_{H,t} - \frac{1-\lambda}{\lambda}p_S e_{S,t} \tag{21}$$

$$\hat{\gamma}_t = \hat{c}_{S,t} - \hat{c}_{H,t} \tag{22}$$

$$\widehat{mc}_t = \hat{w}_t - \hat{a}_t = -\hat{\xi}_t + \varphi\hat{n}_t + \sigma\hat{c}_t + \hat{q}_t - \hat{a}_t \tag{23}$$

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + \frac{\kappa}{\sigma + \varphi}\widehat{mc}_t. \tag{24}$$

- $\psi = \lambda\frac{C_H}{Y}$, $\gamma = \log\Gamma$, $g \equiv \Gamma - 1 > \gamma$, $p \equiv \frac{p_S}{p_S + (1-p_S)\Gamma^\sigma}$, $\eta \equiv \frac{1-\lambda}{\lambda}p_S\eta_S + p_H\eta_H$.
- From (20) if $\Gamma = 1 \Rightarrow \gamma = 0$, $g = 0$, $\lambda = \psi$, then $\hat{q}_t = 0$ (Bilbiie (2025)).

Implications

- Shocks that worsen transition probabilities (lower p_S, higher p_H or both) have independent recessionary demand-side, deflationary supply-side, and worsening distributional effects.
- Optimal monetary policy should overlean against these shocks.
- How much? There is a trade-off.
- Strict inflation targeting is sub-optimal (unlike Bilbiie (2025)).

Partial Equilibrium - Labor Market

- Partial profits redistribution case: $\tau^D = \psi = \lambda \frac{C_H}{Y} < \lambda$.
- Monetary Policy that sterilizes the output gap $x_t = y_t - y_t^* = 0 \, \forall t$.
- Full Log-linear Equilibrium
- We look at the labor market in partial equilibrium.
- Lowercase letters without hats denote log levels.
- Preference shock $\hat{\xi}_t$ that triggers an increase in idiosyncratic risk $\hat{\lambda}_t$.

Partial Equilibrium - Labor Market

Employment

- The equilibrium is:

$$y_t = y_t^* = \frac{1+\varphi}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi} \hat{\xi}_t$$

- Using the production function we get the demand determined employment:

$$n_t = y_t^* - a_t = \frac{1-\sigma}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi} \hat{\xi}_t$$

- Assuming $a_t = a$

$$n_t = y_t^* - a = \underbrace{\left(\frac{1+\varphi}{\sigma+\varphi} a - a\right)}_{y^* - a} + \frac{1}{\sigma+\varphi} \hat{\xi}_t.$$

details

Partial Equilibrium - Labor Market

Labor Supply

- Using the expression for wages we get the aggregate labor supply:

$$w_t = -\hat{\xi}_t + \varphi n_t + \sigma y_t + \hat{q}_t$$

- Assuming $a_t = a$:

$$mrs_t - a = \sigma y^* - a - \frac{\varphi}{\varphi + \sigma}\hat{\xi}_t + \varphi n_t + \hat{q}_t$$

details

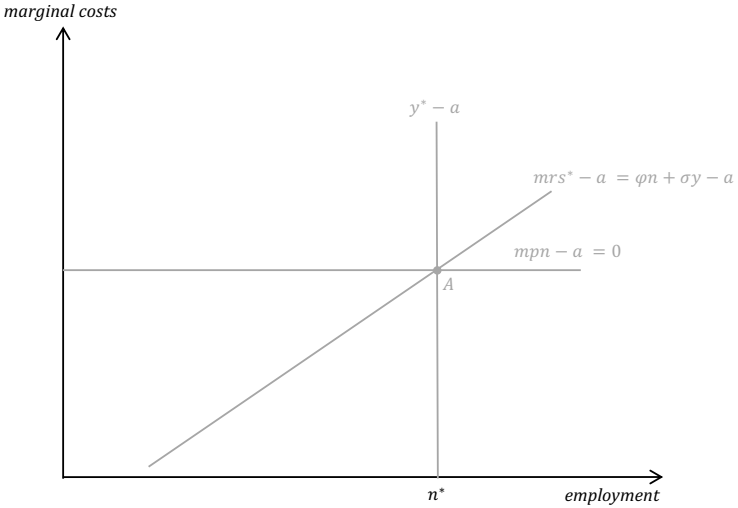
Partial Equilibrium - Labor Market

Equilibrium Markup

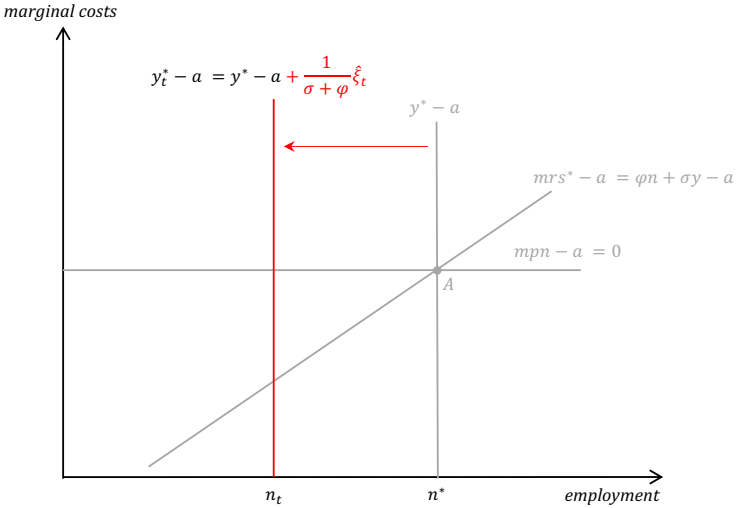
- Substituting the equilibrium values of n_t and y_t in the wage equation:

$$
\begin{aligned}
w_t &= -\hat{\xi}_t + \varphi\left(\frac{1-\sigma}{\sigma+\varphi}a + \frac{1}{\sigma+\varphi}\hat{\xi}_t\right) + \sigma\left(\frac{1+\varphi}{\sigma+\varphi}a + \frac{1}{\sigma+\varphi}\hat{\xi}_t\right) + \hat{q}_t \\
\Rightarrow \mu_{P,t} &= a - w_t = -\hat{q}_t
\end{aligned}
$$

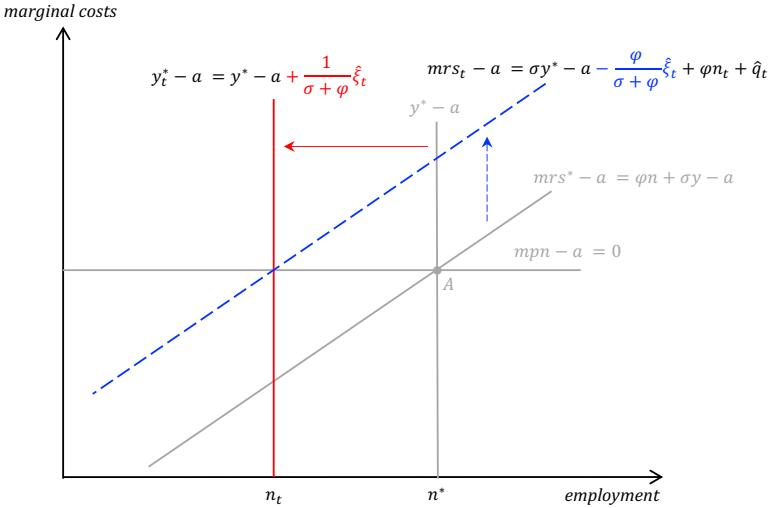
Partial Equilibrium - Labor Market



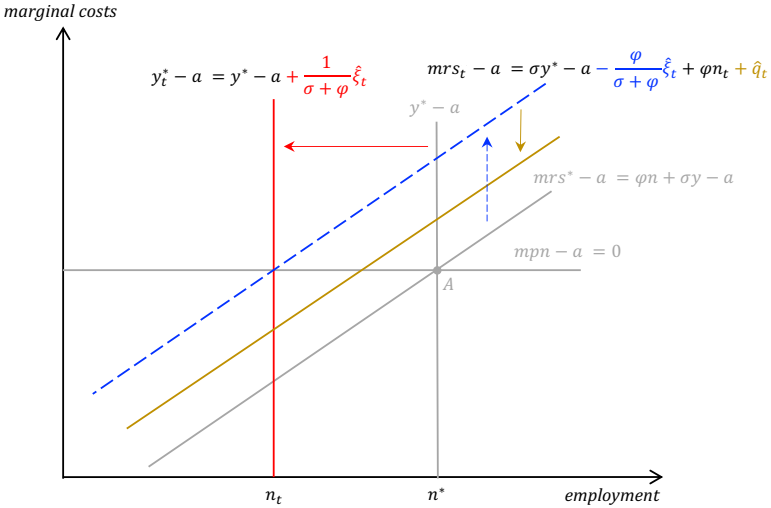
Partial Equilibrium - Labor Market



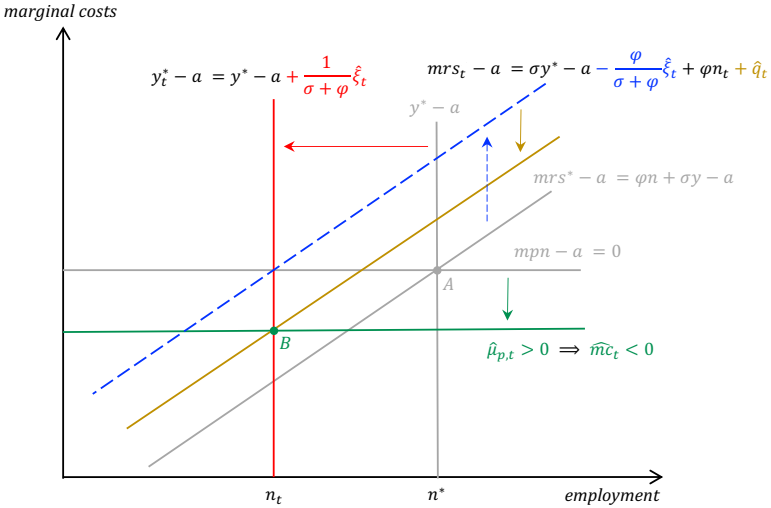
Partial Equilibrium - Labor Market



Partial Equilibrium - Labor Market



Partial Equilibrium - Labor Market



Takeaways

Partial Equilibrium - Labor Market

- Idiosyncratic risk generates a trade-off for monetary policy via its transmission through the extensive margin of $\hat{q}_t$.
- The trade-off is between $\downarrow$ output gap and $\downarrow$ inflation.
- This happens outside of the ZLB.
- Preliminary results with heterogeneity in productivities in the T-HANK model suggest these shocks might also generate stagflation.

Next Steps

- THANK with heterogeneous productivity.
- Quantitative analysis
- Optimal policy

Conclusion

- Household heterogeneity matters on the supply side of HANK.
- Q_t: sufficient-statistic cross-sectional wedge, which feeds into labor supply, marginal costs, and the Phillips curve.
- Decompose Q_t into two relevant margins: intensive and extensive margins of heterogeneity.
- Idiosyncratic risk shocks become recessionary, deflationary, and distributionally adverse.
- Policy should lean harder against these shocks, and strict inflation targeting is not optimal.

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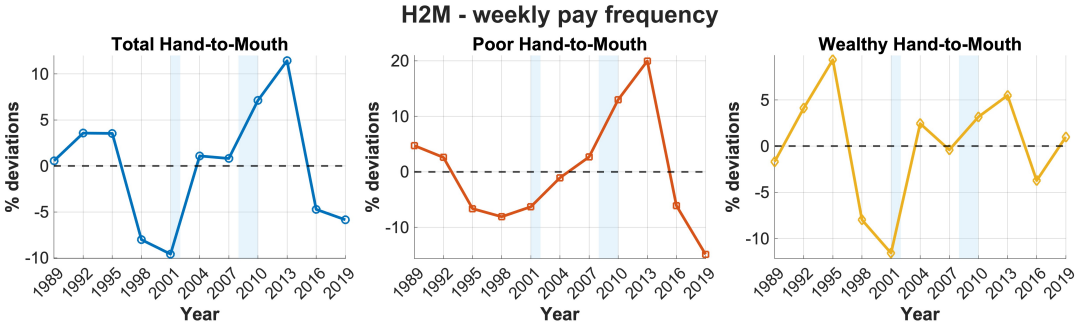
Related Literature

Back

- Labor supply wedge: Berger, Bocola, and Dovis (2023).
- Labor supply & wage setting: Broer et al. (2020), Gerke et al. (2024), Cantore et al. (2025), Graves, Huckfeldt, and Swanson (2023).
- Heterogeneity in consumption baskets & NKPC: Olivi, Sterk, and Xhani (2024), Neyer and Stempel (2025).
- Optimal policy and the divine coincidence in HANK: McKay and Wolf (2025), Dávila and Schaab (2023), Acharya, Challe, and Dogra (2023), Del Negro et al. (2025).

Evolution of HtM

Back



Fluctuations around the linear trend in the proportion of Hand-to-Mouth households, identified as in Kaplan and Violante (2014), from the Survey of Consumer Finances (SCF). A linear time trend has been removed to isolate cyclical variation. The blue shaded areas indicate NBER recessions.

- **Sizable** fluctuations around the trend
- **Countercyclical** H2M share dynamics

Empirical evaluation of $\mathcal{Q}_t$ Back

$$\mathcal{Q}_t = \exp\left(-\frac{1+\varphi}{2\varphi}\sigma_z^2 + \frac{(1+\varphi)\sigma}{\varphi}\rho_{zc,t}\sigma_z\sigma_c - \frac{\sigma(\sigma+\varphi)}{2\varphi}\sigma_c^2\right). \tag{25}$$

- $\mathcal{Q}_t \leq 1$ if and only if the exponent is non-positive.
- Dividing through by $\sigma_z\sigma_c \cdot \sigma(1+\varphi)/\varphi > 0$ and defining $r_t \equiv \sigma_c/\sigma_z$, this reduces to:

$$\rho_{zc,t} \leq \frac{(1+\varphi) + \sigma(\sigma+\varphi)\, r_t^2}{2\sigma(1+\varphi)\, r_t}. \tag{26}$$

Empirical evaluation of $\mathcal{Q}_t$ Back

- Evaluate the implied values for $\mathcal{Q}$ for different calibration of σ and φ, calibrated at $\sigma_z^2 = 0.315$, $\sigma_c^2 = 0.118$, $\rho_{zc} = 0.52$ (Heathcote, Storesletten, and Violante, 2014).

Table: Implied values of $\mathcal{Q}_t$ from equation (25) for alternative calibrations of σ and φ.

$\sigma \setminus \varphi$	0.5	1	2	3	5	10
0.5	0.683	0.772	0.820	0.837	0.851	0.861
1	0.706	0.793	0.840	0.856	0.870	0.880
2	0.631	0.765	0.842	0.870	0.893	0.910
3	0.445	0.656	0.796	0.850	0.895	0.930
5	0.109	0.339	0.596	0.720	0.837	0.938
10	≈0	0.008	0.103	0.239	0.470	0.778

$1 - \mathcal{Q}_t$

Back

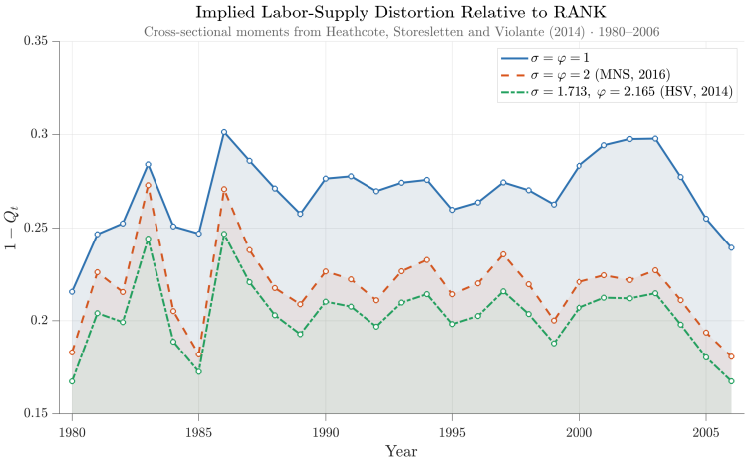


Figure: Fluctuations in the labor-supply distortion relative to RANK, for different calibrations.

HANK Model: Firms

Back

- The final-good firm aggregates differentiated intermediates with CES technology:

$$Y_t = \left(\int_0^1 y_{jt}^{1/\mu} \, dj \right)^{\mu}.$$

- Intermediate firms are monopolistically competitive and produce with linear technology:

$$y_{jt} = A_t n_{jt}.$$

- Price adjustment is costly à la Rotemberg:

$$\psi_t(p_{jt}, p_{jt-1}) = \frac{\mu}{\mu - 1} \frac{1}{2\kappa} \left[\log \left(\frac{p_{jt}}{p_{jt-1}} \right) \right]^2 Y_t.$$

- In symmetric equilibrium, inflation satisfies

$$\log(1 + \pi_t) = \kappa \left(\frac{W_t}{A_t} - \frac{1}{\mu} \right) + \frac{1}{1 + r_{t+1}} \frac{Y_{t+1}}{Y_t} \log(1 + \pi_{t+1}),$$

while dividends are

$$d_t = Y_t - W_t N_t - \psi_t.$$

HANK Model: Policy

Back

- Monetary policy follows a Taylor rule:

$$i_t = r_t^* + \phi \pi_t + \epsilon_t.$$

- The ex-post real rate is pinned down by last period's nominal rate and current inflation:

$$1 + r_t = \frac{1 + i_{t-1}}{1 + \pi_t}.$$

- Fiscal policy supplies a constant stock of public bonds B and uses lump-sum taxes to finance interest payments:

$$\tau_t = r_t B.$$

- So monetary policy governs intertemporal prices, while fiscal policy passively balances the budget given the interest bill on government liquidity.

HANK Model: Market Clearing

Back

- Asset market clearing equates household savings to public bond supply:

$$B = \int_0^1 b_{h,t}\, dh.$$

- Labor market clearing matches labor demand and supply in efficiency units:

$$\frac{Y_t}{A_t} = N_t = \int_0^1 z_{h,t} n_{h,t}\, dh.$$

- Goods market clearing allocates output between consumption and nominal rigidities:

$$Y_t = \underbrace{\sum_{z \in \mathcal{E}} \int c_t(z, b) \mathcal{F}_t(z, b)\, db}_{\text{Aggregate consumption}} + \psi_t.$$

- These conditions close the model: savings equal public liquidity, labor clears in efficiency units, and output is split between consumption and price-adjustment costs.

HHs problem Back

- Utilitarian family head maximising the average lifetime utility across all agents
- Limited risk-sharing: *within* but not *across* types

$$\mathcal{U}_t = \int_0^1 u_{j,t} + \beta E_t \mathcal{U}_{t+1} = (1-\lambda_t) u_{S,t} + \lambda_t u_{H,t} + \beta E_t \mathcal{U}_{t+1} \tag{27}$$

- Labor income is taxed at a rate τ^w. The corresponding revenues are rebated lump-sum pro rata. (efficient allocation in SS even in presence of inequality)
- Subsidy on aggregate employment, $\tau W_t N_t$.
  - Financed by lump-sum taxes on firms (with a share equal to $1-\delta$), and on households (share equal to δ).
  - The share δ is split between hand-to-mouth (HtM) and savers, with hand-to-mouth (HtM) that pays $\lambda_t \delta$ and savers $(1-\lambda_t)\delta$.
- Hand-to-mouth households receive a transfer that redistributes among them a share τ^D of aggregate profits D, financed through a lump-sum tax on savers.

HHs problem

Back

$$C_{S,t}^{-\sigma} = \beta(1+i_t)E_t\left\{\frac{\exp(\Delta\xi_{t+1})}{1+\pi_{t+1}}\left[p_{S,t+1}C_{S,t+1}^{-\sigma} + (1-p_{S,t+1})C_{H,t+1}^{-\sigma}\right]\right\} \tag{28}$$

$$C_{H,t} = (1-\tau^W)W_{H,t}N_{H,t} - T_{H,t} = (1-\delta\tau)W_tN_t + \frac{\tau^D}{\lambda_t}D_t \tag{29}$$

- $T_{H,t} = (\delta\tau - \tau^W)W_{H,t}N_{H,t} - \frac{\tau^D}{\lambda_t}D_t$ denotes the type-specific, per-capita lump-sum taxes net of transfers.

$$N_{j,t}^{\varphi}C_{j,t}^{\sigma}\exp(-\xi_t) = (1-\tau^W)W_{j,t}. \tag{30}$$

Summary of the Model

Back

Condition	Equation
1. Euler equation Savers	$C_{S,t}^{-\sigma} = \beta(1+i_t)E_t\left\{\frac{\exp(\Delta\xi_{t+1})}{1+\pi_{t+1}}\left[p_{S,t+1}C_{S,t+1}^{-\sigma} + (1-p_{S,t+1})C_{H,t+1}^{-\sigma}\right]\right\}$
2. Budget constraint H	$C_{H,t} = W_t N_t + \frac{\tau^D}{\lambda_t} D_t - \delta\tau W_t N_t$
3. Labor supply S	$N_{S,t}^{\varphi} C_{S,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{S,t}$
4. Labor supply H	$N_{H,t}^{\varphi} C_{H,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{H,t}$
5. Marginal cost	$MC_t = (1-\tau)\frac{W_t}{A_t}$
6. Aggregate labor	$N_t = N_{S,t}^{1-\lambda_t} N_{H,t}^{\lambda_t}$
7. Aggregate Wage	$W_t = W_{S,t}^{1-\lambda_t} W_{H,t}^{\lambda_t}$
8. Production function	$Y_t = A_t N_t$
9. Aggregate cons.	$Y_t = (1-\lambda_t)C_{S,t} + \lambda_t C_{H,t}$
10. Profits	$D_t = Y_t(1-MC_t) - (1-\delta)\tau W_t N_t = Y_t - (1-\delta\tau)W_t N_t$
11. Share dynamic	$\lambda_t = \lambda_{t-1} p_{H,t} + (1-\lambda_{t-1})(1-p_{S,t}) = (p_{S,t} + p_{H,t} - 1)\lambda_{t-1} + 1 - p_{S,t}$
12. Monetary policy	$\hat{i}_t = \phi_\pi \hat{\pi}_t + e_{m,t}$
13. Phillips curve	$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{\kappa}{\sigma+\varphi}\widehat{mc}_t$

Aggregate Labor Supply

Back

- From individual labor supplies

$$N_{S,t}^{\varphi} C_{S,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{S,t}, \tag{31}$$

$$N_{H,t}^{\varphi} C_{H,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{H,t}, \tag{32}$$

- Using the Cobb-Douglas labor aggregator $N_t = N_{S,t}^{1-\lambda_t} N_{H,t}^{\lambda_t}$, we get:

$$N_t^{\varphi} = \exp(\xi_t)(1-\tau^w) W_t \left[C_{S,t}^{1-\lambda_t} C_{H,t}^{\lambda_t} \right]^{-\sigma}, \tag{33}$$

$$\tag{34}$$

- Using the production function $Y_t = A_t N_t$, and $C_t = Y_t$ we get the aggregate labor supply:

$$(1-\tau^w) W_t = \exp(-\xi_t) A_t^{-\varphi} C_t^{\sigma+\varphi} Q_t. \tag{35}$$

Steady state

Back

- $\tau = \frac{1}{\epsilon}$, $\tau^W = 1 - Q$, $\mathcal{M} \equiv \frac{\epsilon}{\epsilon - 1}$.
- $D = \frac{\delta}{\epsilon} Y > 0$ and $\tau^D < \lambda$.

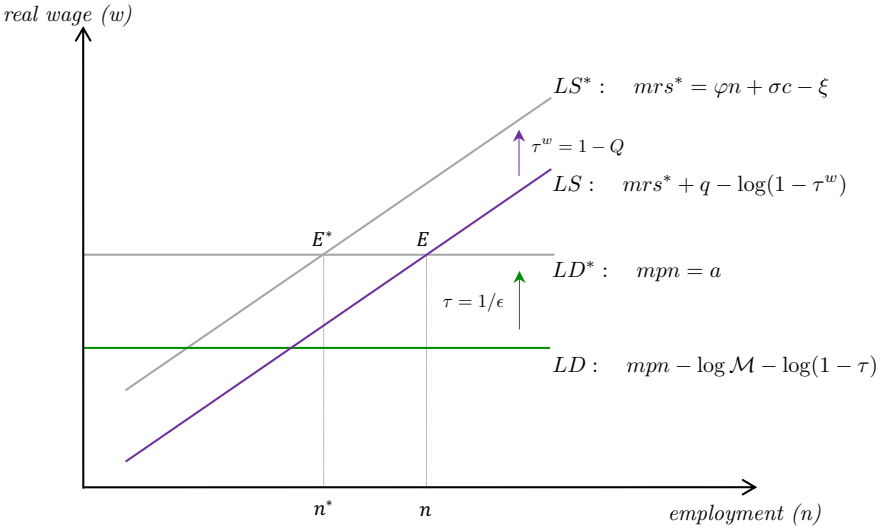
$$\Gamma \equiv \frac{C_S}{C_H} \Rightarrow \Gamma = \frac{1 - \lambda + \lambda \frac{\delta}{\epsilon}\left(1 - \frac{\tau^D}{\lambda}\right)}{1 - \lambda - (1 - \lambda)\frac{\delta}{\epsilon}\left(1 - \frac{\tau^D}{\lambda}\right)} > 1. \tag{36}$$

- Given $\tau^W = 1 - Q$, then $W = N^{\varphi} Y^{\sigma}$. Finally $\Gamma > 1 \Rightarrow Q_t < 1$ which implies that:

$$MC = \frac{1 - \tau}{1 - \tau^W} \frac{Y^{\sigma + \varphi}}{A^{1+\sigma}} Q \Rightarrow Y = A^{\frac{1+\sigma}{\sigma+\varphi}} = Y^*. \tag{37}$$

- Note: without the labor income tax, i.e. $\tau^W = 0$, $Y > Y^*$.

Steady-State



Back

Log-linearized Equilibrium Conditions

Back

Condition	Equation
Euler equation S	$\hat{c}_{S,t} = E_t \hat{c}_{S,t+1} - (1-p) E_t \hat{Y}_{t+1} - \frac{1}{\sigma}\left(\hat{\imath}_t - E_t \hat{\pi}_{t+1} + E_t \Delta \hat{\xi}_{t+1}\right) + p \frac{\Gamma^{\sigma} - 1}{\sigma} E_t \hat{p}_{S,t+1}$
Budget constraint H	$\hat{c}_{H,t} = \left(1 - \frac{\tau^D \delta}{\psi \epsilon}\right)(\hat{w}_t + \hat{n}_t) + \frac{\tau^D}{\psi} \hat{d}_t - \frac{\tau^D \delta}{\psi \epsilon} \hat{\lambda}_t$
Labor supply	$\hat{w}_t = -\hat{\xi}_t + \varphi \hat{n}_t + \sigma \hat{c}_t + \hat{q}_t$
Marginal costs	$\widehat{mc}_t = \hat{w}_t - \hat{a}_t$
Q_t wedge	$\hat{q}_t = \sigma(1-\lambda)\hat{c}_{S,t} + \sigma\lambda \hat{c}_{H,t} - \sigma \hat{y}_t - \sigma\gamma\lambda \hat{\lambda}_t$
Consump. inequality	$\hat{Y}_t = \hat{c}_{S,t} - \hat{c}_{H,t}$
Production function	$\hat{y}_t = \hat{a}_t + \hat{n}_t$
Aggregate consump.	$\hat{c}_t = (1-\psi)\hat{c}_{S,t} + \psi \hat{c}_{H,t} - \psi g \hat{\lambda}_t$

Log-linearized Equilibrium Conditions

Back

Condition	Equation
Resource constraint	$\hat{y}_t = \hat{c}_t$
Profits	$\hat{d}_t = \hat{y}_t - (1-\delta\tau)(\hat{w}_t + \hat{n}_t)$
Share dynamics	$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} - \eta\hat{y}_t + p_H e_{H,t} - \frac{1-\lambda}{\lambda} p_S e_{S,t}$
NK Phillips curve	$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{\kappa}{\sigma+\varphi}\widehat{mc}_t$
Persistence prob. S	$\hat{p}_{S,t} = \eta_S \hat{y}_t + e_{S,t}$
Persistence prob. H	$\hat{p}_{H,t} = -\eta_H \hat{y}_t + e_{H,t}$

Definitions of composite parameters

$$p \equiv \frac{p_S}{p_S + (1-p_S)\Gamma^{\sigma}}, \quad \psi \equiv \frac{\lambda}{\lambda + (1-\lambda)\Gamma}, \quad \rho \equiv p_S + p_H - 1 = 1 - \frac{1-p_S}{\lambda}, \quad \eta \equiv \frac{1-\lambda}{\lambda} p_S \eta_S + p_H \eta_H$$

Log-linear T-HANK

Partial profits redistribution case: $\tau^D = \psi < \lambda$ back

- $\hat{c}_{h,t} = \hat{y}_t + \frac{\delta}{\epsilon}\hat{\lambda}_t \longleftrightarrow \chi = 1$
- We are shutting down the amplification coming from countercyclical consumption inequality.

Description	Equation
IS curve	$x_t = E_t x_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1} - r_t^*) - \left[\theta(1-p\rho) + \rho(1-p)\frac{\delta}{\epsilon}\right]\hat{\lambda}_t + p\frac{\Gamma^{\sigma}-1}{\sigma}E_t\hat{p}_{S,t+1}$
NKPC	$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + \kappa x_t - \kappa\omega\hat{\lambda}_t$
Cons. inequality	$\hat{\gamma}_t = \frac{\varpi}{1-\psi}\hat{\lambda}_t$
Monetary policy	$\hat{i}_t = \phi_\pi\hat{\pi}_t + e_{m,t}$
H share	$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} + p_H e_{H,t} - \frac{1-\lambda}{\lambda}p_S e_{S,t}$
Natural rate	$\hat{r}_t^* \equiv \sigma E_t\Delta\hat{y}_{t+1}^* - E_t\Delta\hat{\xi}_{t+1}$
Potential output	$\hat{y}_t^* = \frac{1+\varphi}{\sigma+\varphi}\hat{a}_t + \frac{1}{\sigma+\varphi}\hat{\xi}_t$

Composite parameters

Composite parameters ($\tau^D = \psi < \lambda$)

Back

Composite parameters
$p_\gamma \equiv p_S + (1-p_S)\Gamma^\sigma$; $p \equiv \frac{p_S}{p_\gamma}$; $\psi \equiv \frac{\lambda}{\lambda+(1-\lambda)\Gamma} = \frac{\lambda}{1+g(1-\lambda)}$; $\chi = 1$; $\phi = 0$; $\Phi = 1$; $\sigma_x = \sigma$; $\omega_x = 0$; $\varpi = \frac{\delta}{\epsilon} + \psi g$; $\zeta = \psi\left(g + \frac{\delta}{\epsilon}\right)$; $\theta = \frac{\psi}{1-\psi}\left(g + \frac{\delta}{\epsilon}\right)$. $\nu = \frac{\psi\left(g+\frac{\delta}{\epsilon}\right)-(1-p)\left(\frac{\delta}{\epsilon}+\psi g\right)}{1-\psi}$; $\omega = \frac{\sigma}{\sigma+\varphi}\left[\frac{\lambda-\psi}{1-\psi}\left(\frac{\delta}{\epsilon} + \psi g\right) + \lambda\gamma - \psi g\right]$; $\rho = 1 - \frac{1-p_S}{\lambda} \in (0, 1)$.

Partial Equilibrium - Labor Market back

Employment

- Start from:

$$y_t^* = \frac{1+\varphi}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi}\hat{\xi}_t$$

so

$$y_t^* - a = \frac{1+\varphi}{\sigma+\varphi} a_t - a + \frac{1}{\sigma+\varphi}\hat{\xi}_t.$$

- Add and subtract $\frac{1+\varphi}{\sigma+\varphi}a$:

$$y_t^* - a = \underbrace{\left(\frac{1+\varphi}{\sigma+\varphi}a - a\right)}_{y^*-a} + \frac{1+\varphi}{\sigma+\varphi}(a_t - a) + \frac{1}{\sigma+\varphi}\hat{\xi}_t.$$

- Assumption needed for the simplified expression: $a_t = a$ (no productivity deviation). Then

$$y_t^* - a = y^* - a + \frac{1}{\sigma+\varphi}\hat{\xi}_t, \qquad y^* - a = \frac{1-\sigma}{\sigma+\varphi}a.$$

Partial Equilibrium - Labor Market

Labor Supply back

- Start from the wage equation and subtract a:

$$w_t - a = -\hat{\xi}_t + \varphi n_t + \sigma y_t^* - a + \hat{q}_t$$

- Using

$$y_t^* = y^* + \frac{1}{\sigma + \varphi}\hat{\xi}_t$$

we can rewrite

$$-\hat{\xi}_t + \sigma y_t^* = \sigma y^* + \frac{\sigma}{\sigma + \varphi}\hat{\xi}_t - \hat{\xi}_t = \sigma y^* - \frac{\varphi}{\sigma + \varphi}\hat{\xi}_t$$

- Therefore,

$$w_t - a = \sigma y^* - a - \frac{\varphi}{\sigma + \varphi}\hat{\xi}_t + \varphi n_t + \hat{q}_t$$